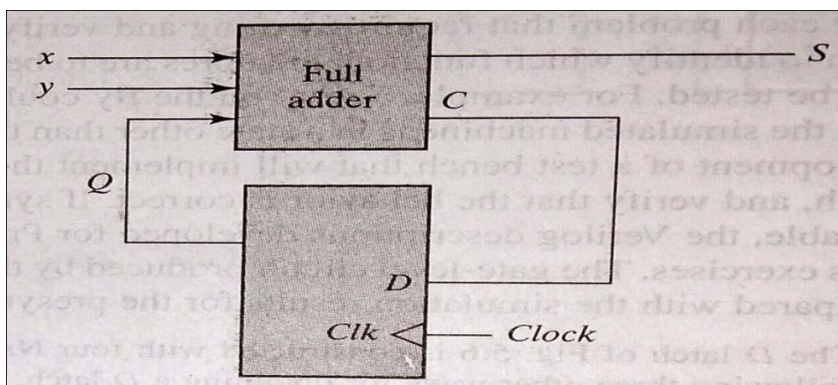


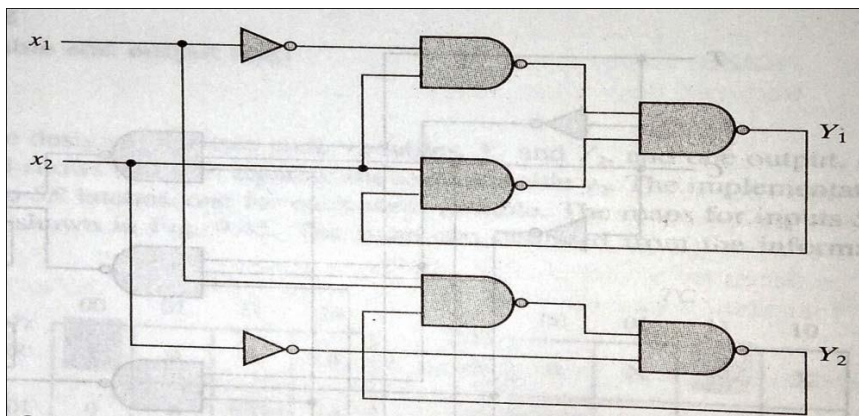
Question Bank

UIT		BRANCH-CSE	Even Semester 2023-24	SEMESTER:4 th	
		SUBJECT-DIGITAL ELECTRONICS		SUBJECT CODE-BOE310	
UNIT- 1			CO-1		
SECTION –A (SHORT ANSWER TYPE QUESTIONS)				Blooms Taxonomy Level	Session
1	A	Interpret the binary number (1011) ₂ into (i) Gray code (ii) Excess-3Code		K1	2022-23
	B	Evaluate (1011) ₂ - (1101) ₂ using 1’s and 2’s complement method.		K1	2022-23
	C	Explain the signed binary number.		K1	2022-23
	D	Define the term universal gates and their applications.		K1	2022-23
	E	Define the term binary codes with an example.		K1	2121-22
	F	Differentiate between SOP & POS form.		K1	2121-22
	G	How are binary digits used to express the integer and fractional parts of a number?		K1	2121-22
	H	Explain how BCD addition is carried out.		K1	2121-22
SECTION –B (LONG ANSWER TYPE QUESTIONS)					
2	A	Implement the Boolean function F (x, y, z) = (1,2,3,4,6,7) using NAND gates.		K2	2022-23
	B	Simplify Y=∑m (3,6,7,8,10,12,14) + d (0,1,15) using K-map method and implement the simplified circuit using logic gates.		K2	2022-23
	C	Minimize the following Boolean function using tabulation method: F (A, B, C, D, E) = ∑ m (0,4,12,16,19,24,27,28,29,31)		K2	2022-23
	D	Convert the following, i) (5162) ₁₀ = () ₂ ii) (11011001) ₂ = () ₁₀ iii) (6273) ₁₀ = () ₈ iv) (7860) ₁₀ = () ₁₆ v) (A23B8) ₁₆ = () ₁₀		K2	2022-23
	E	Design an XOR gate by using NAND gate implementation.		K2	2022-23
	F	Define the De-morgans theorem of Logic Simplification for SOP & POS forms.		K2	2022-23
	G	Explain the implementation of an X-OR gatewithNANDimplementation.		K2	2022-23
	H	ExplaindifferentstepsassociatedtoQuineMcClusky(TabularMethod)ofminimizing BooleanFunctions.		K2	2121-22
	I	MinimizethefollowingBooleanfunctionusingKMap f (A,B,C,D)=∑m (0,1,4,8,9,10)+ ∑d (2,11)		K2	2121-22
	J	Realize a 3-input gate using 2-input gates for the following gates: (i) AND (ii) OR (iii) NAND (iv) NOR		K2	2121-22
	K	Minimize the following using Tabularmethod F (A, B, C, D, E)=∑m (0,1,2,3,6,7,14,15,16,19,31)		K3	2121-22
	L	(i) Reducetheexpressionf=∑m(0,1,2,3,5,7,8,9,10,12,13)usingK-mapsandimplementthe real minimal expression using NAND logic. (ii) Designthelogiccircuit foraBCD todecimaldecoder.		K2	2121-22

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		SUBJECT-DIGITAL ELECTRONICS		SUBJECT CODE-BOE310		
UNIT- 2			CO-2			
SECTION –A (SHORT ANSWER TYPE QUESTIONS)				Blooms Taxonomy Level	Session	
1	A	Differentiate between the serial and parallel adder.		K1	2022-23	
	B	How many 4×1 multiplexers are required to implement 64 ×1 multiplexer?		K1	2022-23	
	C	Elaborate the term Combinational Circuits.		K1	2022-23	
	D	Define BCD codes and convert (A5D8) ₁₆ into BCD number.		K1	2022-23	
	E	Define the term combinational logic with an example.		K1	2121-22	
	F	Discuss universal gates.		K1	2121-22	
	G	Implement a 4:1 multiplexer using 2:1 multiplexer.		K1	2121-22	
	H	Demultiplexer is decoder circuit with an additional enabling input. Do you agree with the above statement?		K1	2121-22	
SECTION –B (LONG ANSWER TYPE QUESTIONS)						
2	A	Construct a full adder and implement the full adder with the help of half adders. Also implement the full adder with NAND gates only.			K2	2022-23
	B	Design a BCD adder using 4-bit parallel adder.			K2	2022-23
	C	Draw and Explain 2-bit magnitude comparator. Also represent output with the help of logic diagram.			K2	2022-23
	D	Design 4:1 multiplexer using gates			K2	2022-23
	E	Design a 4-bit adder circuit using gates.			K2	2022-23
	F	Design a 3:8 Decoder circuit using gates.			K2	2022-23
	G	Illustrate the working of Serial and parallel adders and differentiate the operations.			K2	2022-23
	H	Design a 4-bit magnitude comparator			K2	2121-22
	I	Design a full adder and full subtractor using NAND gates only.			K2	2121-22
	J	Implement a full subtractor circuit using only NAND gates.			K2	2121-22
	K	Using 4:1 multiplexer, implement the following function $F(A, B, C) = \sum m(0,2,3,5,7)$			K2	2121-22
	L	Explain 4-bit magnitude comparator.			K2	2121-22

UIT		BRANCH-CSE	Even Semester 2023-24	SEMESTER:4 th	
		SUBJECT-DIGITAL ELECTRONICS		SUBJECT CODE-BOE310	
UNIT- 3			CO-3		
SECTION –A (SHORT ANSWER TYPE QUESTIONS)				Blooms Taxonomy Level	Session
1	A	What is the difference between characteristic and excitation table.		K1	2022-23
	B	Differentiate between combinational and sequential circuits.		K1	2022-23
	C	Explain the term storage elements.		K1	2022-23
	D	Illustrate the term sequential logic.		K1	2022-23
	E	Explain the term Latch		K1	2121-22
	F	Explain the term registers.		K1	2121-22
	G	Give the difference between positive and negative edge triggering.		K1	2121-22
	H	A flip-flop has 5 ns delay from the time the clock edge occurs to the time the output is complemented. What is the maximum delay in a 10-bit binary ripple counter that uses these flip-flops? What is the maximum frequency the counter can operate reliably?		K1	2121-22
SECTION –B (LONG ANSWER TYPE QUESTIONS)					
2	A	Discuss excitation table for SR, JK, T and D flip flop.		K2	2022-23
	B	Design and implement MOD-10 synchronous counter.		K2	2022-23
	C	For the clocked JK Flip-Flop write the state table, state equation with state diagram.		K2	2022-23
	D	Elaborate the characteristic equations of S-R and J-K Flip-Flops.		K2	2022-23
	E	Elaborate the working and circuit of a Serial-in-Serial-Out shift register.		K2	2022-23
	F	Explain the working and circuit of a modulo-5 counter using gates.		K2	2022-23
	G	Explain the working of J-K Flip-Flop.		K2	2022-23
	H	Describe the Design of J-K FF using T FF.		K2	2121-22
	I	Describe the operations and applications of a Serial-in Parallel-out shift register with a neat diagram.		K2	2121-22
	J	Define bi-directional shift register. Draw and explain 3-bit bi-directional shift register using D flip-flop.		K2	2121-22
	K	Design a 3-bit UP/DOWN counter with a direction control M, using JK flip-flops.		K3	2121-22
	L	A sequential circuit has one flip-flop Q, two inputs x and y, and one output S. It consists of a full-adder circuit connected to a D flip-flop, as shown in given figure. Derive the state table and state diagram of the sequential circuit. What does this circuit represent?		K3	2121-22



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		SUBJECT-DIGITAL ELECTRONICS		SUBJECT CODE-BOE310	
UNIT- 4			CO-4		
SECTION –A (SHORT ANSWER TYPE QUESTIONS)				Blooms Taxonomy Level	Session
1	A	Define race around condition in JK flip flop.		K1	2022-23
	B	Define term propagation delay.		K1	2022-23
	C	Define the use of clock in digital circuits.		K1	2022-23
	D	Explain the term synchronous circuits.		K1	2022-23
	E	Define Asynchronous circuits.		K1	2121-22
	F	Discuss hazards.		K1	2121-22
	G	Define critical race and non-critical race.		K1	2121-22
	H	What is the significance of state assignment?		K1	2121-22
SECTION –B (LONG ANSWER TYPE QUESTIONS)					
2	A	Discuss Mealy and Moore finite state machine with an example.		K2	2022-23
	B	Explain State Reduction and assignment with suitable example.		K2	2022-23
	C	Design a sequential circuit with two flip flops, A & B and one input X. When X=0 state of the circuit remains the same, when X =1 circuit passes through the state transition from 00 to 01 to 11 to 10 back to 00 and repeat.		K2	2022-23
	D	Illustrate the State reduction technique for Digital Circuits.		K2	2022-23
	E	Illustrate the working and applications of Asynchronous sequential circuits.		K2	2022-23
	F	Explain the term, Hazard. Define different types of Hazards along with detection and reduction of Hazards.		K2	2022-23
	G	Define the state reduction steps for a machine.		K2	2022-23
	H	Design a sequential circuit with two flip flops A & B and one input x. when x = 0, the state of the circuit remains the same and when x = 1 the circuit passes through the state transitions from 00 to 01 to 11 to 10 back to 00 and repeat.		K2	2121-22
I	A sequential circuit has two J K flip flops A & B, two inputs X & Y, and one output Z. The equations defining this system are as following: $JA = BX + BY' \quad KA = B'XY' \quad JB = A'X \quad KB = A + XY'$ $Z = AXY + BX'Y'$ Design the circuit.		K3	2121-22	
J	Design a primitive state diagram and state table for a circuit with two asynchronous inputs (X and Y) and one output Z. This circuit is to be designed so that if any change takes place on X and Y, Z is to change states. Assume initially that the two inputs never change simultaneously.		K3	2121-22	
K	Derive the transition table for the asynchronous sequential circuit shown in given figure. Determine the sequence of internal states Y ₁ Y ₂ for the following sequence of inputs X ₁ X ₂ : 00,10,11,01,11,10,00.		K3	2121-22	
					
L	(i) Distinguish between static and dynamic hazard. How will you determine hazard in combinational circuits? (ii) Draw the logic diagram of the product-of-sum expression $Y = (X_1 + X_2')(X_2 + X_3)$ Show that there is a static 0-hazard when X ₁ and X ₃ are equal to 0 and X ₂ goes from 0 to 1. Find a way to remove the hazard by adding one more OR gate.		K3	2121-22	

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		SUBJECT-DIGITAL ELECTRONICS		SUBJECT CODE-BOE310	
UNIT- 5			CO-5		
SECTION –A (SHORT ANSWER TYPE QUESTIONS)				Blooms Taxonomy Level	Session
1	A	Give the difference between PAL and PLA.		K1	2022-23
	B	How many address lines are needed to represent 8K memory.		K1	2022-23
	C	Illustrate the use of logic families in digital circuits.		K1	2022-23
	D	Elaborate the term Fan-in in digital circuits.		K1	2022-23
	E	Discuss logic family and its use.		K1	2121-22
	F	What do you mean by a memory?		K1	2121-22
	G	Why is ECL logic faster than TTL?		K1	2121-22
	H	Compare static RAM and dynamic RAM.		K1	2121-22
SECTION –B (LONG ANSWER TYPE QUESTIONS)					
2	A	Design 8K x 8 RAM memory system, using 1Kx8 memory ICs.		K2	2022-23
	B	Why ECL is better? Implement NAND gate with DTL and TTL.		K2	2022-23
	C	Define noise margin, Fan-in, Fan-out as characteristics of logic families. Implement NAND gate with CMOS.		K2	2022-23
	D	Define the TTL (Transistor-Transistor-Logic) logic Family used for digital circuits.		K2	2022-23
	E	Define the SRAM cell with working and circuit diagram along with applications.		K2	2022-23
	F	Elaborate the PLA (Programmable Logic Array) along with working and applications.		K2	2022-23
	G	Discuss different types of RAM memory cell.		K2	2022-23
	H	Explain the working and structure of EEPROM cell.		K2	2121-22
	I	Describe the difference between PAL & PLA using neat diagram and suitable examples.		K2	2121-22
	J	(i) Write a note on interfacing TTL with CMOS. (ii) Explain the parameters used to characterize logic families.		K2	2121-22
	K	Design a BCD to Excess-3 code converter and implement it using a suitable PLA.		K2	2121-22
	L	Draw a neat diagram of TTL NAND gate and explain its operation.		K2	2121-22